Lecture 14: Absolutely Convergence; The Ratio and Root Test

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Based on Thomas' Calculus, 15th Edition in SI Units, Pearson Education Limited

Sem II, 2025/2026

Learning Outcome

At the end of this lecture, you should be able to:

- ☐ Understand what is absolutely convergence and able to identify if a series is converges absolutely.
- ☐ Use the Ratio Test to tell whether a series converges or diverges when possible and know the limitations of the test.
- ☐ Use the Root Test to tell whether a series converges or diverges when possible and know the limitations of the test.

Absolute Convergence

A series $\sum a_n$ converges absolutely (or is absolutely convergent) if the corresponding series of absolute value, $\sum |a_n|$ converges.

The Absolute Convergence Test

Absolutely Convergence Test

If $\sum_{n=1}^{\infty} |a_n|$ converges, then $\sum_{n=1}^{\infty} a_n$ converges.

The Absolute Convergence Test

Example: $\sum_{n=1}^{\infty}(-1)^{n+1}\frac{1}{n^2}$

The Absolute Convergence Test

Example: $\displaystyle\sum_{n=1}^{\infty} \frac{\sin n}{n^2}$

The Absolute Convergence Test

The converse is not necessarily true.

Example: $\sum_{n=1}^{\infty}(-1)^n \frac{1}{n}$

Ratio Test

The Ratio Test

Let $\sum a_n$ be any series and suppose that

$$\lim_{n\to\infty} \left| \frac{a_{n+1}}{a_n} \right| = \rho.$$

Then

- the series converges absolutely if $\rho < 1$,
- the series diverges if $\rho > 1$ or ρ is infinite,
- the test is inconclusive if $\rho = 1$.

Ratio Test

Ratio Test

Example: $\sum_{n=0}^{\infty} \frac{2^n + 5}{3^n}$

Ratio Test

Example: $\sum_{n=1}^{\infty} \frac{(2n)!}{n!n!}$

Ratio Test

Example: $\sum_{n=1}^{\infty} \frac{4^n n! n!}{(2n)!}$

Ratio Test

Example: $\sum_{n=0}^{\infty} \frac{n^2 2^n}{3^n}$

Root Test

Consider the series with the terms

$$a_n = \begin{cases} n/2^n, & n \text{ odd} \\ 1/2^n, & n \text{ even.} \end{cases}$$

What happens if we apply ratio test to the series?

Root Test

The Root Test

Let $\sum a_n$ be any series and suppose that

$$\lim_{n\to\infty} \sqrt[n]{|a_n|} = \rho.$$

Then

- the series converges absolutely if $\rho < 1$,
- the series diverges if $\rho > 1$,
- the test is inconclusive if $\rho = 1$.

Root Test

Back to the series with the terms

$$a_n = \begin{cases} n/2^n, & n \text{ odd} \\ 1/2^n, & n \text{ even.} \end{cases}$$

Root Test

Example: $\sum_{n=1}^{\infty} \frac{n^2}{2^n}$

Root Test

Example: $\sum_{n=1}^{\infty} \frac{2^n}{n^3}$

Root Test

Example: $\sum_{n=1}^{\infty} \frac{1}{1+n}$

Root Test

Example: $\sum_{n=1}^{\infty}(1-\frac{1}{n})^{n^2}$